

SECTION 4 – SITE ENVIRONMENT

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4-1 INTRODUCTION

The rating and duty cycles of all subsystems are applicable only if the room environment is maintained as specified in the following sections. The environment must be constantly maintained (i.e. holidays, weekends, etc.) to prevent exceeding these restrictions. Subjecting the equipment to consistent excessive temperatures and humidity above specifications may shorten the life of the internal electrical components.

4-2 TEMPERATURE AND HUMIDITY SPECIFICATIONS

Note

The temperature and humidity specifications for Signa MR/i & CV/i Upgrades are not changed. However, if these specifications in the equipment room are not strictly adhered to, failures of the SGD Gradient Cabinet modules may occur.

Use the specifications listed in Table 4-1 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmental controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

To help prevent a patient from feeling uncomfortably warm during a scan, ensure that the magnet room temperature does not exceed 70°F (21.1°C) maximum.

TABLE 4-1
TEMPERATURE AND HUMIDITY SPECIFICATIONS

AREA	TEMPERATURE RANGE °F (°C)	TEMPERATURE REGULATION °F (°C)	TEMPERATURE CHANGE °F/Hr (°C/Hr)	HUMIDITY (%)	HUMIDITY CHANGE (%/Hr)	MAX. ROOM GRADIENT °F (°C)
Inlet to Computer/ Equipment	50–70* (10–21.1) 65 (18.3) desired temperature	± 5 (3) within temperature range	5 (3)	30–60* See Note 1	5	5 (3)**
Magnet Room	≤ 70 (21.1)	per patient comfort	5 (3)	30–60*	5	5 (3)
Operator's Control Room or Independent Console Room	68–78* (20–25) 72 (22.2) desired for operator comfort	per operator comfort	5 (3)	30–60*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65°F (18.3°C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets. 1 Equipment reliability is severely impacted if equipment is exposed to high humidity. High voltage circuits in SGD Gradient Cabinet will breakdown and fail if condensation forms.						

4-3 COOLING REQUIREMENTS

4-3-1 Air Cooling

Table 4-2 contains the heat output of the upgrade equipment listed in the typical site location. These values do not include people, lights, magnet equipment cabinets, and non-MR equipment. Actual site average values will vary depending on system use (ie. protocols used, patient load, etc.). Note any variations of equipment location for your site when calculating your cooling requirements for each room.

Requirements

Due to the large variation in Equipment Room heat loads, the air conditioner compressor should be equipped with unloaders or hot gas bypass to prevent moisture stripping of the evaporator coils.

The Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Enclosure Rear Pedestal, location of coil/bore cooling fans intake.

Recommendations

A dedicated air conditioner with a dual compressor is preferred to avert shutdowns during repair of the primary air conditioner.

It is recommended that a temperature and humidity recorder be used during preinstallation and during actual installation and placed near the Gradient Cabinet air inlets to establish the true criteria. Refer to cooling table calculator in this section for each room's cooling requirements.

GE recommends the use of a 12 inch high raised flooring system for the equipment room (10 inch minimum clearance from floor slab to underside of access flooring). Care must be taken in locating the air conditioning supply vents in the floor. The air conditioning supply vents should be located directly in front of the cabinet inlets.

TABLE 4-2
MAXIMUM MR AIR COOLING TABLE

MR COMPONENT	1.5T SIGNA MR/i OR CV/i MAX HEAT OUTPUT†						1.0T SIGNA MR/i MAX HEAT OUTPUT†					
	MAGNET ROOM (SEE NOTE 2)		EQUIPMENT ROOM		OPERATOR/ CONTROL AREA		MAGNET ROOM (SEE NOTE 2)		EQUIPMENT ROOM		OPERATOR/ CONTROL AREA	
	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT	BTU/HR	WATT
RF/Gradient Body Coil Assembly, Magnet Enclosure Equipment	12,285	3600					9215	2700				
Blower Box	3415	1000					3415	1000				
Compact PDU			7000	2050					Not Applicable			
RF/Penetration Cab			10,235	3000					Not Applicable			
RF/PDU Cab			Not Applicable						10,235	3000		
SGD (See Note 1) – Horizon LX SGD – Horizon LX SGD HiSpeed – Horizon LX SGD EchoSpeed			10,925 27,645 27,645	3200 8100 8100					10,925 27,645 N/A	3200 8100 N/A		
System Cabinet			6140	1800					6140	1800		
Water Chiller for Gradient Coil			14,000	4100					14,000	4100		
Magnet Monitor			205	60					205	60		
Shield/Cryo Cooler Compressor			See Note 3						See Note 3			
Operator Workspace (See Note 4) with – B/W Monitor or – Color Monitor or – LCD Color Monitor					4710 5050 4095	1380 1480 1200					4710 5050 4095	1380 1480 1200
*Main Disconnect (See Note 5)			900	264					900	264		
*ADW Cabinet Signa CV/i Option			225	75					Not Applicable			
*ADW Color Monitor Signa CV/i Option					427	125					Not Applicable	

Note * Optional equipment.

† Maximum heat output is defined for temperature and humidity as defined in Table 4-1.

†† This component may be located in other areas of the site.

- Gradient Driver Equipment consists of one of the following:

Base	SGD Base Cabinet	For 1.0T & 1.5T Signa MR/i
HiSpeed	SGD Hi-Slew Cabinet	For 1.0T & 1.5T Signa MR/i
EchoSpeed	SGD Hi-Slew Cabinet	For 1.5T Signa MR/i & CV/i
- Magnet Room must be an individual temperature zone controlled by a separate thermostat to allow for adjustments to meet room temperature specification as listed in Section 4-2, TEMPERATURE AND HUMIDITY SPECIFICATIONS. It is recommended that cool inlet air be directed towards the Blower Box intake which contains coil/patient cooling fans.
- Shield Cooler/Cryo Compressor Cabinet requires water cooling to dissipate heat output. Heat dissipation to air is negligible.
- Operator Workspace equipment includes the following: Monitor (B/W or Color), Octane Computer, Workspace Cabinet, Mouse and Mouse Pad, LCD Panel, Keyboard, and interface modules mounted to Workspace Table
- Values are for Main Disconnect Control option available from GEMS. Heat output may be different if customer supplied control is used.

4-3-2 Water Cooling

Gradient Coil For Fixed Sites

The Gradient Coil is air and water cooled. A water-to-air Water Chiller (WC1) is provided with the fixed site system. De-ionized water and anti-freeze mixture is supplied with the Gradient Coil for use in the closed loop system.

Note

The Water Chiller is a dedicated closed loop unit for Gradient Coil water cooling **only** and not to be shared with other equipment, refer to Section 4-3-3 Water Chiller (WC1) For Gradient Coil Cooling.

Note

The Gradient Coil has GE provided flexible hose (0.5 in. (12.7 mm) inside diameter) and adjustable compression clamps for connections. Refer to Section 6, INTERCONNECT DATA, for flexible hose routing information.

Gradient Coil For Mobiles

Mobile systems (True Mobile, Transportable, Relocatable) utilize a water-to-water heat exchanger for Gradient Coil Cooling which replaces the air-to-water Heat Exchanger (HE1) used in the Signa Horizon system. The Mobile van will provide chilled water for the new heat exchanger operation.

Shield Cooler Compressor

Magnet equipment (Shield/Cryo Cooler) water cooling requirements are unchanged.